

5(a)	<i>three from:</i> any four named structures ; mouth / nose / nostrils / nasal passages (count as one structure) pharynx larynx trachea A windpipe bronchus / bronchi bronchiole / bronchioles alveolus / alveoli / alveolar ducts / alveolar sacs lungs trachea, branches / divides, into (two main) bronchi ; correct sequence from bronchus (branching) to bronchioles to (end with) alveoli ; further detail ; ; e.g. trachea has, C-shaped / incomplete rings, of cartilage bronchus has cartilage, plates / AW diameters of gas exchange structures / respiratory tubes, decrease (towards alveoli)	3
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Question	Answer	Marks
5(b)	<i>two from:</i> blood pressure, decreases / returns to normal ; platelets no longer sticky / risk of thrombosis decreases / risk of blood clotting decreases ; endothelial repair (more likely to be) successful ; A endothelium no longer damaged A decreased risk of, atheroma / atherosclerosis I less fatty deposits heart rate, decreases / returns to normal ; I heartbeat decreases AVP ; e.g. increased peripheral circulation / AW blood vessels, not / less, constricted A less vasoconstriction A blood vessel (lumen) diameter increases / vasodilation occurs	2

Question	Answer	Marks
5(c)	<i>four from:</i> <i>assume from point of view of giving up smoking unless stated otherwise</i> <i>in lungs / at high partial pressures of oxygen</i> 1 (so) more oxygen taken up by haemoglobin / haemoglobin more saturated (with oxygen) / AW ; A haemoglobin has, increased / normal, affinity for oxygen 2 before and after data / calculated difference in data, to support, from 10 kPa ; 3 carbon monoxide no longer binding to haemoglobin / AW ; A carboxyhaemoglobin no longer forming / ora with ref. to smoking A haemoglobin has a higher affinity for CO than oxygen 4 more (sites on) haemoglobin available to take up oxygen ; <i>in respiring tissues</i> 5 more oxygen dissociates from (oxy)haemoglobin / AW, at, lower partial pressures / below 4 kPa / in (respiring) tissues (after stopping) ; A haemoglobin less saturated (with oxygen) A haemoglobin has, less / decreased, affinity for oxygen 6 before and after data / calculated difference in data, to support ; 7 ref. to increase in allosteric effect / regains ability for conformational change / AW ; <i>health benefits after stopping</i> 8 more / sufficient, oxygen to, transport / deliver to, tissues / body or more oxygen to reach (respiring) tissues ; A less, tired / fatigued A more able to, do work / carry out exercise / be mobile R more energy unless qualified with link to aerobic respiration	4

Question	Answer	Marks
5(d)	<i>two from:</i> <i>non-infectious</i> not from a pathogen ; A bacteria / viruses / fungi <i>for pathogens</i> A from lifestyle A tar / carcinogen in smoke, causes mutation (in cells of lung) cannot be passed onto another person / not transmissible / AW ; <i>disease</i> abnormal condition (affecting an organism) / condition that reduces the effectiveness of the functions of the organism / AW ;	2

Question	Answer	Marks
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